# Infosys - Coding 14

HackerRank

You are given 3 integers n, x, k where n is divisible by k. Your task is to construct an array a of size n such that:

• the product of each segment of length k in a is equal to x,

• the elements of a fall between 2 and x ($2 \leq a[i] \leq x$),

• the array a is lexicographically minimal.

Define $p = 10^9 + 9$. Find the sum of (a[i] * p^i) modulo $10^9 + 7$, considering indexing starts from 0. If no answer exists, print -1.

**Input Format**

First line: integer n, the number of elements in a.

Second line: integer x, the condition integer.

Third line: integer k, the length of the segment.

**Constraints**

1 <= n <= 10^6

1 <= x <= 10^6

1 <= k <= 10^6

(n is divisible by k)

**Output Format**

The output contains an integer value

**Sample Input 0**

```
2
2
2
```

**Sample Output 0**

```
-1
```

**Explanation 0**

n=2, x=2, k=2. A length-2 segment needs two elements ≥ 2 whose product is 2, but the minimum product is 2×2 = 4. x=2 has only one prime factor (Ω=1 < k=2), so no array can be constructed → −1.

### Sample Input 1

```
2
2
1
```

### Sample Output 1

```
6
```

### Explanation 1

=2, x=2, k=1. Each single-element segment must have product 2, so every element is 2: array [2, 2]. Sum = $2 \cdot p^0 + 2 \cdot p^1 \equiv 2 + 4 = 6 \pmod{10^9+7}$. [2, 2] is the optimal array.

### Sample Input 2

```
4
10
2
```

### Sample Output 2

```
60
```

### Explanation 2

n=4, x=10, k=2. Lex-min factorization of 10 into two factors ≥ 2 is [2, 5], repeated to length 4: [2, 5, 2, 5]. Sum = $2 \cdot p^0 + 5 \cdot p^1 + 2 \cdot p^2 + 5 \cdot p^3 \equiv 2 + 10 + 8 + 40 = 60 \pmod{10^9+7}$. [2, 5, 2, 5] is the optimal array.